

YUCATAN BASIN BUOY 42056, Mexico

WMO: 997395

Lat: 19.820N

Lon: 84.945W

Elev: 0

StdP: 14.70

Time Zone: -6.00 (W06)

Period: 05-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	71.4	72.7	55.2	64.9	72.4	57.1	69.5	73.5	32.4	73.9	29.6	75.3	20.0	40	0.647

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	2.9	86.3	78.8	86.1	78.8	85.7	78.8	81.0	84.4	80.7	84.5	80.1	84.4	10.5	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.0	155.7	83.8	79.4	152.6	83.8	79.0	150.3	83.8	44.6	84.3	44.2	84.4	43.8	84.4	84.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
28.5	25.9	23.9	69.7	87.0	1.7	1.3	68.5	88.0	67.5	88.7	66.6	89.5	65.3	90.4		
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	80.8	77.6	78.0	78.1	79.9	81.3	82.7	83.4	84.1	83.7	82.4	80.0	78.8
	DBStd	2.86	2.29	2.00	1.99	1.48	1.21	1.33	1.14	1.22	1.40	1.65	1.87	2.26
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11257	856	783	870	897	970	980	1037	1056	1011	1003	900	894
	CDD65	5782	391	363	405	447	505	530	572	591	561	538	450	429
	CDH74	62903	2922	2835	3155	4379	5590	6498	7347	7828	7346	6559	4581	3863
	CDH80	16639	68	29	53	413	1221	2219	2900	3376	3054	2183	753	370
Wind	WSAvg	14.7	16.7	15.8	15.8	15.5	14.1	14.7	13.3	12.0	11.8	14.7	16.6	16.0
Precipitation	PrecAvg	34.5	1.8	1.0	0.8	0.5	2.4	4.1	2.2	3.0	3.7	7.0	5.5	2.3
	PrecMax	52.8	5.0	3.3	1.9	1.9	8.7	9.2	6.0	6.3	9.5	20.4	15.6	8.1
	PrecMin	18.0	0.5	0.2	0.2	0.0	0.2	0.2	0.4	0.7	0.8	1.0	1.0	0.5
	PrecStd	9.9	1.2	0.7	0.5	0.4	2.5	2.6	1.6	1.6	2.5	5.2	3.5	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	81.8	81.0	81.6	83.6	84.0	85.5	86.3	86.5	87.0	86.2	84.5	83.3
		MCWB	75.1	74.8	75.1	79.2	77.8	79.2	79.3	79.0	78.0	78.9	78.3	77.1
	2%	DB	81.2	80.9	81.0	82.8	83.6	85.3	85.8	86.3	86.3	85.5	84.1	82.7
		MCWB	74.8	74.7	74.9	78.0	77.6	79.2	79.1	79.0	78.4	78.6	78.0	76.8
	5%	DB	80.8	80.6	80.7	82.3	83.3	84.8	85.4	86.1	86.1	85.2	83.5	82.4
		MCWB	74.7	74.5	74.8	77.3	77.5	79.1	79.0	78.9	78.6	78.5	77.6	76.7
	10%	DB	80.4	80.1	80.3	81.8	82.9	84.5	85.2	85.8	85.7	84.7	83.0	81.8
		MCWB	74.5	74.2	74.5	76.6	77.2	79.0	79.0	78.8	78.6	78.3	77.2	76.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.0	78.9	77.8	81.0	80.0	81.7	81.0	81.0	81.6	80.7	80.0	78.8
		MCDB	79.2	78.9	80.2	83.1	82.8	84.0	84.3	85.2	84.5	84.7	83.2	82.0
	2%	WB	78.0	77.3	77.0	79.9	79.2	81.0	80.8	80.6	80.7	80.0	79.2	78.2
		MCDB	79.2	79.8	80.1	82.7	82.7	83.9	84.5	85.2	84.8	84.4	83.2	81.9
	5%	WB	76.8	76.0	76.3	78.3	78.8	80.5	80.2	80.1	80.1	79.7	78.8	77.7
		MCDB	79.3	79.9	80.1	82.0	82.6	83.9	84.5	85.1	84.8	84.3	83.1	81.9
	10%	WB	75.8	75.3	75.6	77.4	78.3	80.0	79.9	79.8	79.7	79.2	78.1	77.1
		MCDB	79.5	79.8	80.1	81.4	82.4	83.8	84.5	85.1	84.7	84.1	82.8	81.8
Mean Daily Temperature Range	5% DB	MDBR	2.4	2.1	2.0	1.8	2.1	2.4	2.6	2.9	3.1	3.2	2.9	2.5
		MCDBR	2.8	2.2	2.3	2.1	2.3	2.5	2.8	2.9	3.4	3.2	3.5	3.0
		MCWBR	2.7	2.7	2.6	2.2	2.3	2.5	2.8	2.8	2.9	2.9	3.0	2.9
	5% WB	MCDBR	2.4	2.8	2.3	2.0	2.5	2.5	2.8	3.0	3.3	3.2	3.6	2.7
		MCWBR	2.9	3.3	3.1	2.7	2.6	2.7	2.9	3.0	3.2	3.1	3.1	3.0
Clear-Sky Solar Irradiance	taub	0.379	0.384	0.384	0.428	0.447	0.475	0.489	0.447	0.420	0.408	0.392	0.387	
	taud	2.550	2.528	2.523	2.374	2.368	2.316	2.253	2.416	2.473	2.531	2.535	2.545	
	Ebn at Noon	285	290	293	279	271	261	258	270	278	279	279	279	
	Edh at Noon	30	33	34	40	40	41	44	38	35	32	31	30	
All-Sky Solar Radiation	RadAvg	1378	1695	1977	2178	2103	1900	2106	2022	1829	1537	1403	1308	
	RadStd	95	96	65	76	163	144	129	90	136	178	88	76	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page